

[Full Name]

Senior Full-Stack AI Engineer | Python & Automation Specialist [Email] | [Phone] | [LinkedIn URL] | [Portfolio/GitHub]

Professional Summary

Hands-on Senior Software Engineer specializing in building robust AI-driven applications and automation ecosystems. Expert in Python (FastAPI/Pydantic) development, 3D web visualization, and creating complex system integrations. Passionate about writing clean, efficient code and building tools that bridge the gap between advanced AI models and practical user interfaces.

Technical Skills

- **Backend Development:** Python 3.10+, FastAPI, Pydantic, RESTful APIs, WebSockets, Asyncio.
- **Frontend & Visualization:** JavaScript (ES6+), WebGL, Three.js, HTML5/CSS3, Mobile Responsive Design.
- **AI Integration:** LangChain, MCP (Model Context Protocol) Implementation, OpenAI/Gemini API integration, Local LLM setup.
- **DevOps & Tools:** Git, Docker Compose, Linux Shell Scripting (Bash/Zsh), GitHub Actions (CI/CD), Home Assistant API.
- **Databases:** PostgreSQL, Redis, Vector Databases (Chroma/Milvus).

Key Projects & Experience

Unified System Core Development | *Lead Developer*

Built the core software components for a unified smart home and AI platform.

- **"Vasya-Online" AI Gateway:** Developed a high-performance FastAPI gateway serving as the central nervous system for AI operations. Implemented strict Pydantic validation for all data flows and successfully integrated Google's Gemini Pro Vision for multimodal analysis.
- **Interactive 3D Visualization:** Built a responsive, mobile-friendly 3D infrastructure visualizer using WebGL/Three.js to render real-time system topology, server status, and network connections.
- **Protocol Development:** Standardized communication using the Model Context Protocol (MCP), enabling seamless interoperability between different AI agents and tools.

Automation & Scripting

- **System Health Automation:** Wrote comprehensive Python and Bash scripts for automated reliable backup (Rsync), system health checks, and service recovery (Systemd watchdogs).
- **Home Assistant Integration:** Developed custom Python integrations for Home Assistant to control legacy hardware via generic TCP/HTTP bridges, significantly extending hardware lifecycle.
- **Self-Healing Mechanisms:** Implemented "Ring of Resilience" scripts that automatically failover DNS and connectivity settings to maintain network uptime during ISP outages.

Infrastructure Management

- **Containerization:** Containerized all microservices using Docker Compose, optimizing resource usage on limited hardware (NUC/Raspberry Pi).
- **Hardware Debugging:** Diagnosed and resolved low-level hardware conflicts (USB pass-through, GPU driver issues) in a virtualized Windows/Linux environment.

Languages

- **English:** Professional Proficiency
- **Hebrew:** Professional Proficiency
- **Russian:** Native